Royal Devon and Exeter NHS Foundation Trust

Guidance for management of **Anti-Rheumatic Drugs in the perioperative period**

INTRODUCTION

This guidance has been jointly written by the Department of Rheumatology and Pharmacy for use by surgical teams and Preparation for Surgery Practitioners.

Please consult the Rheumatology Team and/or the surgeon /anaesthetist if there are any concerns.

PUBLICATION DETAILS

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In consultation with	Rheumatology MDT
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SUMMARY OF MAIN GUIDANCE

cDMARDs: CONTINUE these medications through surgery	Dosing Interval	Continue / Withhold
Methotrexate	Weekly	Continue
Sulfasalazine	Once or twice daily	Continue
Hydroxychloroquine	Once or twice daily	Continue
Leflunomide	Daily	Continue
Apremilast	Daily	Continue
Biologic Agents: See main text. Aim to schedule surgery at end or just beyond dosing cycle (column 3). RESUME at minimum 14 days after surgery in the absence of wound healing problems, surgical site infection or other infection	**Dosing Interval**	**Principle to miss 1 dose and schedule surgery (relative to last biologic agent dose administered) during:**
Adalimumab (Humira/Imraldi)	Weekly or every 2 weeks	Week 3
Etanercept (Benepali/Enbrel)	Weekly or twice weekly	Week 2
Golimumab (Simponi)	Every 4 weeks (SC)	Week 5-6
Infliximab (Remicade/Inflectra)	Every 4, 6 or 8 weeks	Week 5-6,7-8 or 9-10
Abatacept (Orencia)	Monthly (IV) weekly (SC)	Week 5-6, Week 2
Certolizumab (Cimzia)	Every 2 weeks Every 4 weeks	Week 3-4 Week 5-6
Rituximab (Mabthera/Truxima/Rixathon)	Every 6 (or more) months IV	Month 3 onwards
Tociluzimab (Ro-Actemra)	Every week (SC) Monthly (IV)	Week 2-3 Week 5-6
Sarilumab (Kevzara)	Every 2 weeks (SC)	Week 2-3
Anakinra (Kineret)	Daily SC	Day 2-3
Secukinumab (Cosentyx)	Every 4 weeks SC	Week 5-6
Ixekizumab (Taltz)	Every 4 weeks SC	Week 5-6
Brodalimuab (Kyntheum)	Every 2 weeks SC	Week 2-3
Ustekinumab (Stelara)	Every 12 weeks SC	Week 13-15
Guselkumab (Tremfya)	Every 8 weeks SC	Week 7-8 onwards
Belimumab (Benlysta)	Every 4 weeks IV	Week 5-6
Tofacitinib (Xeljanz)	Daily or Twice daily orally	7 days after last dose
Baricitinib (olumiant)	Daily orally	7 days after last dose
Filgotinib (Jyseleca)	Daily orally	7 days after last dose
Upadacitinib (Rinvoq)	Daily orally	7 days after last dose
Belimumab (Benlysta)	Every 4 weeks IV	After week 4 but discuss with rheumatology first please
Other immunosuppressive agents used for conditions including SLE, vasculitis and other connective tissue diseases	**Dosing interval**	**Comments**

Cyclophosphamide	IV every 2-4 weeks, occasionally daily oral	Treatment implies active SLE/vasculitis. Need to discuss with renal/rheumatology team
Mycophenolate	Twice daily oral	These medications can be continued through the peri-operative period. However, if the underlying condition is mild or stable they may be withheld for 1 week prior to surgery. Please contact rheumatologist for advice as unplanned withdrawal may be dangerous.
Azathioprine	Daily oral	
Cyclosporin	Daily oral	
Tacrolimus	Twice daily oral	

EXPANDED GUIDANCE

1. Non-steroidal anti-inflammatory drugs (NSAIDs)

- Aspirin –See 'Clinical Guideline for: Aspirin, Clopidogrel and Dipyridamole'.
- Conventional non selective NSAIDs. These drugs have antiplatelet effect that is reversible once the drug is out of the system ie approximately 5 half lives. Ibuprofen can be stopped 1 day before surgery, naproxen 4 days. Some NSAIDs have long elimination half-lives e.g. piroxicam 50hrs (stop 10 days pre-op and tenoxicam 72hrs (stop 15 days pre-op).
- Selective Cox II inhibitor NSAIDs. These drugs lack a significant effect on platelet function and therefore don't have to be withheld for this reason
- Consider withdrawing NSAIDs perioperatively in patients with impaired renal function and who may require other nephrotoxic drugs.

2. Glucocorticoids[1]

- Normal cortisol secretion is 20-30mg per day and a major surgical stress may require 200mg per day. A suitable regimen for corticosteroid replacement, in patients who have taken more than 5mg prednisolone daily (or equivalent) for 4 weeks or longer, within 3 months of surgery, is:
- *Intraoperative management*
- Any surgery under general anaesthesia – usual oral corticosteroid dose on the morning of surgery. Hydrocortisone 100 mg intravenously on induction, followed by immediate initiation of a continuous infusion of hydrocortisone 200 mg.24 h^{-1}.
- Alternatively IV dexamethasone 6-8mg will suffice for 24 hours
- *Postoperative management*
  - Hydrocortisone 200 mg.24 h−1 by IV infusion while nil by mouth or for patients with postoperative vomiting (alternatively, hydrocortisone 50 mg every 6 h by IM injection)
  - Resume enteral glucocorticoid at –double hydrocortisone doses for 48 h if recovery is uncomplicated or for up to a week following major surgery.

3. Conventional Disease modifying anti-rheumatic drugs (DMARDs)

- A few days off a DMARD during a period of illness or perioperatively does not and affect the long term control of rheumatoid arthritis or other immune diseases.
- Please do not stop DMARD drugs and then omit to tell patients when to restart as this <u>will</u> have an adverse effect on overall disease control.
- Stop immunosuppressive treatment if significant renal impairment, other major organ dysfunction, septicemia or bone marrow toxicity such as neutropenia.

a. Methotrexate

- For routine surgery, continue with Methotrexate therapy during the perioperative period (RCT evidence suggests less complications in elective orthopaedic surgery if continued).
- This drug is given <u>weekly</u>, orally or subcutaneously, and adjusting the day of administration by a day or two is acceptable.
- Consider stopping preoperatively if significant renal impairment or other major organ dysfunction.

b. Leflunomide

- This drug has bowel elimination after initial renal clearance and has an extremely long half-life.

- Elimination protocols are complicated and require a 2 week course of activated charcoal or colestyramine.
- If there is an indication for withdrawing leflunomide pre-surgery please contact rheumatologist in good time.

c. Other DMARDs such as hydroxychloroquine, sulfasalazine, gold, penicillamine and azathioprine

- Please continue these drugs perioperatively unless significant renal impairment or other major organ dysfunction and discuss with rheumatologists if necessary

d. Immunosuppression in patients with systemic immune disease, such as systemic lupus erythematosus (SLE & vasculitis

- If patients on IV cyclophosphamide - liaise with renal/rheumatology team; could delay infusions though current treatment implies active vasculitis/SLE
- Patients are often taking hydroxychloroquine, azathioprine and mycophenolate - for these oral maintenance medications please also liaise with the team, as unplanned withdrawal may be dangerous

4. Biologic and other new targeted therapies

The evidence is mixed whether biologic therapy has an adverse effect on surgical outcome. Some studies have shown a small increase in infection risk but others have not. However, the current precautionary policy is to stop biologic treatment so blood and tissue levels are low prior surgery – please see below or liaise with rheumatology team for exact arrangements. Some surgical teams are happy for a patient to continue with their treatment if the procedure is deemed very low risk e.g. minor hand surgery, removal of metalwork.

All biologic therapy can be restarted approximately 2-3 weeks after surgery when incisions have healed and the surgical team is happy there is no infection.
For intravenous therapy (all Rituximab & infliximab, some Tocilizumab & Abatacept) – please see specific guidance below and liaise with rheumatology team regarding the timing of surgery to enable infusions to be scheduled accordingly.

a. Anti-TNFα therapy [Etanercept (*Benepali or Enbrel*); Adalimumab (*Humira or Imraldi*); Infliximab (*Remicade or Inflectra*); Certolizumab pegol (*Cimzia*) & Golimumab (*Simponi*)]

- *Intravenous Anti-TNFα therapy:* Infliximab is an intravenous infusion with an induction regime (at times 0, 2 weeks and 6 weeks) and then every 6-8 weeks. Aim for surgery toward end of infusion cycle IE at week 6-8. Discuss with rheumatology re: infusion scheduling
- *Subcutaneous Anti-TNFα therapy:* miss at least 1 injection prior to surgery
  - Etanercept injection 50mg s.c. once a week or 25mg twice weekly
  - Adalimumab injection 40mg s.c. injection every 2 weeks.
  - Certolizumab pegol 200mg s.c. every 2 weeks.
  - Golimumab 50-100mg s.c. monthly
- Can be restarted approximately 2-3 weeks after surgery when incisions have healed and the surgical team is happy there is no infection.

b. Rituximab (MabThera, anti-B cell therapy)

- Rituximab is an intravenous infusion used to treat rheumatoid arthritis and vasculitis. Each course consists of a single or two infusions given two weeks

apart, administered on the day case unit. Courses can be repeated after a minimum of 6 months.

- Ideally plan surgery at least 3 months (ideally six-months) after most recent infusion. Discuss with rheumatology if surgery is required earlier than this.

c. Tocilizumab (*Ro-Actemra)* and Sarilumab (*Kevzara*); both anti-Il-6 therapy

- Tocilizumab & sarilumab are used to treat rheumatoid arthritis and Giant Cell Arteritis.
- Tocilizumab is administered either intravenously once every four weeks or weekly as a subcutaneous injection. Sarilumab is a subcutaneous injection every 2 weeks.
- For intravenous tocilizumab, aim for surgery toward end of infusion cycle. Discuss with rheumatology re: infusion scheduling
- For subcutaneous tocilizumab & sarilumab, miss at least 1 injection prior to surgery.
- Can be restarted approximately 2-3 weeks after surgery when incisions have healed and the surgical team is happy there is no infection.

d. Ustekinumab (Stelara) and guselkumab (Tremfya); both anti-Il-12/23 pathway

- Ustekinumab and guselkumab are subcutaneous injections used to treat psoriatic arthritis.
- Ustekinumab is administered every 3 months. Surgery can be planned around doses; towards the end of the 3 month period.
- Guselkumab is administered every 8 weeks. Surgery can be planned around doses; towards the end of the 8 week period

e. Secukinimab (Cosentyx), Ixekizumab (Taltz) and brodalumab (Kyntheum); anti-Il-17 therapy

- Secukinimab, ixekizumab and brodalumab are used to treat psoriasis, psoriatic arthritis and ankylosing spondylitis
- Mostly administered monthly as a subcutaneous injection, though brodalumab is fortnightly
- Miss at least 1 injection prior to surgery
- Can be restarted approximately 2-3 weeks after surgery when incisions have healed and the surgical team is happy there is no infection.

f. Abatacept (*Orencia*; prevents full T-cell activation)

- Abatacept is used to treat rheumatoid arthritis (rarely I.V.)
- Given as a weekly subcutaneous injection.
- Miss at least 1 injection prior to surgery
- Can be restarted approximately 2-3 weeks after surgery when incisions have healed and the surgical team is happy there is no infection.

g. JAK inhibitors: tofacitinib (Xeljanz), baricitinib (Olumiant), filgotinib (Jyseleca) and upadacitinib (Rinvoq)

- Given orally, to treat rheumatoid arthritis

- Tofacitinib 5mg once - twice daily, baricitinib 2-4mg once daily, Filgotinib 100-200 mg daily and upadacitinib 15 mg daily
- Short half-life; stop 1 week before surgery
- Can re-start 1-2 weeks after surgery once surgical team happy there is no infection.

h. Apremilast (Otezla, inhibits phosphodiesterase type 4)

- Given orally to treat psoriatic arthritis
- Maintenance dose of 30mg BD.
- Can be treated in same way as conventional DMARD with respect to perioperative period (i.e. can be continued)

i. Belimumab (Benlysta, BLyS inhibitor)

- Used in systemic lupus erythematosus (SLE) to reduce B-cell survival and production of autoantibodies
- Administered intravenously every 4 weeks
- Half-life 19-20 days
- Surgery best planned at week 4 but by the very nature of the SLE patients on this drug (complicated/severe disease), best to discuss with rheumatology team please
- Can be restarted approximately 2-3 weeks after surgery when incisions have healed and the surgical team is happy there is no infection

j. Anakinra (Kineret; anti-Il1ra)

- Anakinra 50-100mg daily subcutaneously is used to treat rheumatoid arthritis
- Short half-life; stop 1 week before surgery
- Can be restarted approximately 2-3 weeks after surgery when incisions have healed and the surgical team is happy there is no infection.

5. Relevant contacts

Rheumatology Nurse Specialist Team: Jill Moran, Tracey Morey, Sarah Wellington, Augustina Makasy and Katy Lewis

- Epic MyCare: Rheumatology nurse staff message pool: **p RH8 OPC M Rheumatology – nurse**
- extension 3502.
- Rheumatologists via secretary, switchboard (registrar bleep) or available on Consultant Advice Line 07920 781631 between 12:30 and 13:30pm Monday to Friday.
  - Consultants: Richard Haigh, Ravik Mascarenhas, Susie Earl, Matthew Cates, Roberta Rabbitts & Mahdi Abusalameh
  - Staff grade: Danny Murphy
  - Specialist doctor: Daniel Bartram rheumatology
- Rheum SpR on bleep
- T, O & R Directorate Pharmacist: Faye Kester

Reference:

Guidelines for the management of glucocorticoids during the peri-operative period for patients with adrenal insufficiency
Guidelines from the Association of Anaesthetists, the Royal College of Physicians and the Society for Endocrinology UK
Available at: Guidelines for the management of glucocorticoids during the peri-operative period for patients with adrenal insufficiency - Woodcock - 2020 - Anaesthesia - Wiley Online Library